

Sahaana B

Coimbatore, Tamil Nadu, India | sahaana.john@gmail.com | +91 81220 85655
linkedin.com/in/-sahaana-b-/ | sahaana-balan.github.io/

Executive Summary

Final Year Mechatronics Engineering student with strong analytical, problem-solving, and technical skills in CAD design, automation, and electromechanical systems. Hands-on experience in design validation and engineering projects, with a passion for applying technology-driven solutions in industrial and digital environments.

Technical Skills

- **Programming & Technical Skills:** C++, Python, Data Structures & Algorithms(DSA), Object-Oriented Programming(OOP), Basics of SQL, HTML, CSS
- **CAD & Engineering Tools:** SolidWorks, AutoCAD, CAD Modelling, Mechanical Design, Simulation & Analysis

Education

Sri Krishna College of Engineering and Technology B.E. Mechatronics Engineering CGPA: 8.1/10.0	2023 – 2027
Sri Sowdeswari Vidhyalaya Matric Higher Secondary School Higher Secondary Percentage: 88	2022 – 2023

Projects

Solar Electric Car - Team ASTRA

Team Project | Role: Steering Head

- Optimized the steering subsystem geometry and linkage using SolidWorks and CAD simulations for stable vehicle handling.
- Strengthened skills in mechanical design, fabrication, testing, and cross-functional teamwork, contributing to **1st Place at the Bharat Solar Vehicle Challenge (Oct 2025)**.

Assistive Feeding and Medication Dispensing Robot

Group Mini Project | Mechatronics Engineering

- Developed the mechatronic system architecture and 3D CAD layouts to validate mechanical feasibility.
- Built practical knowledge in system integration, motion analysis, and interdisciplinary collaboration through integrating feeding and dispensing mechanisms.

Internship Experience

Design Intern

May 2026

VVN Engineering, Coimbatore, Tamil Nadu

- Developed & modified 3D CAD models and technical drawings to support product design and manufacturing.
- Acquired practical insight into manufacturing workflows, design validation, and industrial engineering practices.

Intern – Power Plant Operations & Industrial Systems

May 2025

NLC India Limited, Neyveli, Tamil Nadu

- Observed industrial operations across mechanical, electrical, and control systems in power plant environment.
- Learned preventive maintenance practices, industrial safety standards, and large-scale operational workflows.

Certifications & Technical Workshops

- Introduction to Machine Learning - NPTEL (IIT Kharagpur)
- Mastering Collaborative Robots Workshop - PSG College of Technology, Coimbatore
- Wheeled Mobile Robots - NPTEL (IIT Madras)
- Additive Manufactured Bio-Implants Workshop - Indian Institute of Technology, Palakkad